

Sai Siddharth Cilamkoti

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EDUCATION

University at Buffalo

Master of Science in Computer Science

Buffalo, New York
Aug. 2023 -- Feb 2025

- Concentration: Machine Learning and Artificial Intelligence
- Relevant Courses: Data Structures & Algorithms, Operating Systems, Machine Learning, Deep Learning

Amrita Vishwa Vidyapeetham

Bachelor of Technology in Computer Science with AI

India
Aug. 2019 -- May 2023

TECHNICAL SKILLS

Programming: Python, Java, SQL, Javascript, C++, HTML/CSS

AI & ML: PyTorch, LangChain, LangGraph, VectorDBs, Sklearn, NumPy, Huggingface, Fine-Tuning, Prompt Engineering, RAG, LangFuse, LangSmith, AWS Bedrock, Gemini, Anthropic, OpenAI API

Backend & API: Spring Boot, FastAPI, Flask, WebSockets, Microservices

DevOps and Tools: Docker, GitHub Actions, AWS, Azure Dev Tools, Git, Linux, Cursor

EXPERIENCE

Associate AI Engineer

Midwest Partners LLC

Nov 2025 -- Present
Remote

- Owned an end-to-end data pipeline using n8n, SerpAPI (Google Hotels API), and Supabase (PostgreSQL), designing scalable SQL data models and schemas to monitor competitor hotel rates in real time and surface cost-strategy insights.
- Built a scheduling and alerting workflow to monitor business processes and system outputs in real time — detecting when client pricing exceeded competitors and firing Slack/email notifications tied to Google metasearch ranking signals to drive actionable recommendations.
- Integrated the Google Places API for review monitoring and sentiment scoring across Google, TripAdvisor, and OTA platforms, structuring the data into business-intelligence dashboards for accurate, traceable information delivery.

Research AI Engineer

University at Buffalo | Co-authored a [paper](#) on this study, accepted at FLLM 2025

Feb. 2024 -- Aug 2025
Buffalo, New York

- Designed and built a large-scale LLM evaluation pipeline benchmarking 20 frontier models (GPT-4o, Claude 3.5 Sonnet, DeepSeek-v3, Gemini, Llama, Grok) across 878 coding tasks with 100+ test cases each, using OpenAI, Anthropic, Google Vertex AI (GCP), and AWS Bedrock APIs with batch inference and iterative prompt-feedback loops that lifted Pass@25 from 51% to 92%.
- Built an automated, agentic LLM-as-judge review pipeline that scored generated solutions and ran 25-iteration regeneration on execution errors, producing quality signals that flagged DeepSeek-v3 and GPT-4o as most efficient while Gemini-1.5-Pro ran up to 450x less efficiently on Dynamic Programming and Backtracking tasks.
- Optimized a transformer-based model (GPT-2) for text summarization via knowledge distillation, 8-bit quantization, and structured pruning, cutting inference energy 30% while retaining 95%+ of baseline ROUGE quality on the evaluation benchmark.

Software Engineer

Kaleida Health

Aug. 2024 -- Dec. 2024
Buffalo, New York

- Owned an end-to-end, full-stack feature spanning a React frontend, Java/Spring Boot backend services, and scalable microservices, deploying to AWS cloud infrastructure with a focus on production-grade availability and reliability.
- Built real-time, streaming WebSocket messaging services with 1:1 and group chat across care teams and auto-translation via the Google Translate API, securing all API endpoints with JWT, HTTPS, and RBAC across patient, family, doctor, and admin roles.
- Implemented patient-delegated access enforcing correct, role-scoped data access, and drove the backend SDLC and documentation across cross-functional teams to ship the stakeholder-demoed platform.

PROJECTS

FrontDesk - Agentic Natural Language Query System | Python, FastAPI, SQLite, Streamlit, OpenAI API, Langfuse, sqlglot

[GitHub](#) | 2026

- Designed and shipped a production tiered-hybrid NL-to-SQL agent over a hotel dataset, using OpenAI function-calling to route queries across a parameterized tool, a semantic query-spec compiler, and a guarded text-to-SQL fallback, minimizing LLM autonomy while preserving open ended question coverage and eliminating single-point-of-failure failure modes.
- Built an end-to-end LLM reliability stack with sqlglot AST-level validation, read-only execution, one-shot self-correction retries, grounded narration constrained to retrieved rows, and per-stage tracing with evaluation harnesses over a routing/SQL test suite to measure path selection accuracy, hallucination rate, and decline correctness using langfuse.

InlineAsk - Browser Extension | TypeScript, Chrome MV3, Anthropic & OpenAI APIs

[GitHub](#) | 2025

- Built and shipped an end-to-end Chrome extension in TypeScript enabling inline LLM follow-up questions on highlighted text across Claude, ChatGPT, Gemini, and Copilot, with a pluggable multi-provider layer integrating the Anthropic and OpenAI APIs.
- Engineered frontend DOM adapters applying context-engineering discipline — retrieval-style context grounding under a strict 4000-token budget — with AES-GCM credential encryption and retry/timeout handling to harden edge cases and ensure reliability.